

Human-Agent Dialogues and their Purposes

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ABSTRACT

A common conversation between an older adult and a nurse about health-related issues includes topics such as troubles with sleep, reasons for walking around nighttime, pain conditions, etc. Such a dialogue can be regarded as a "natural" dialogue emerging from the participating agents' lines of thinking, their roles, needs and motives, while switching between topics as the dialogue unfolds. The purpose of this work is to define a generic conceptual model of purposeful human-agent dialogue activity including different types of argumentation dialogues, suitable for health-related topics. This is done based on analyses of a scenario, persona and models of human behaviour. The model will be shared between the human and the agent, allowing for adaptation to the human's reasoning, needs and motives.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative interaction** ; • **Computing methodologies** → *Intelligent agents*; Discourse, dialogue and pragmatics;

KEYWORDS

Human-agent interaction, Intention recognition, Dialogue systems, Formal argumentation, End-user development, Activity theory, Ontologies, Multi-agent systems, Conversational interfaces

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1 INTRODUCTION

Our work focuses on dialogues between a human and an *active* assistive technology [8] in the form of an intelligent software agent. A software agent is defined as a computer system that is situated in some environment and that is capable of autonomous action in this environment to meet its design

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objectives [16]. From a cognitive ergonomics perspective, with the development of new interfaces, new ways of interaction are required. Erik Hollnagel [4] states that the design of HCI must include a comprehensive task analysis and dialogue design. Embodied Cognitive Agents (ECA) such systems [1] uses a virtual representation of a human with the ability to send information through body language in addition to linguistic messages. However, we restrict the focus in this work to structured linguistic dialogues based on semantic models of relevant knowledge similar to Hunter in [5, 6].

The goal is to build a software agent that interacts with the human as their personal coach, friend or a discussion partner, i.e., as a *Coach Agent* as described in [11]. In this work, we focus on the dialogue activity, and restrict the agent's autonomy to the agent's task of selecting topics, dialogue types and moves within a dialogue initiated by the human.

In particular, we are interested in how a dialogue, which includes the types of argumentation dialogues information-seeking, inquiry (generate new knowledge), deliberation (deciding about what actions to do) and persuasion dialogues defined by Walton and Krabbe [14] can be organized following theories of purposeful human activity (Activity Theory [7] and Self-Determination Theory [13]). The main contribution of this paper is a conceptual model of the Coach Agent's knowledge regarding conducting the dialogue activity for assessment purposes.

The paper is organised as follows. A description of the methods applied is provided in the following section. The scenario that was used for defining the model is presented in Sect. 3 and a generic model of the dialogue activity is provided in Sect. 4. The article ends with some conclusions and directions for future work.

2 METHODS

The semantic model for adaptive human-agent dialogues is designed based on the persona and scenario of a female older adult named Eva, and the dialogues aimed for supporting a human actor described in [10, 11]. The scenario was analyzed, providing baseline requirements and a conceptual model of goal-directed dialogue activity. The theoretical base for analysis is Activity Theory [7], which also informed the models generated as results of our work.

3 MODEL BASED ON A SCENARIO

Our persona called Eva shares similarity with some of the participants in a study conducted by Lindgren and Nilsson [9], and is therefore considered representative. Eva is 84 years old, suffers from pain in her back and legs and had suffered from few falls before a hip fracture. We envision that Eva

begins to walk around nighttime, and that she may discuss the situation and her sleep with a nurse. The nurse asks a few specific questions about Eva’s activities and health, and this dialogue will wander from one aspect to another, sometimes coming back to a topic already mentioned.

This example of a natural dialogue is rather different from dialogues described in literature, which aims at reaching one particular goal of a dialogue, e.g., [2]. Following the categorization of dialogue types, described by Walton and Krabbe [14], the dialogue with the nurse is a simplified example of a combination of different goals: finding information (*information-seeking* type), generating new knowledge, i.e., conclusions (*inquiry dialogue* type) and deciding upon actions to make (*deliberation* type). In case one of the agents has reasons for arguing for one particular action to be made with the purpose to convince the other, e.g. for safety reasons, the dialogue may include a *persuasive* part, e.g., to convince Eva that she needs to go to the hospital for investigation. Based on this, we can define a set of *generic goals* for the agent to use in its organization of dialogues. More concretely, the generic outcome of each type of dialogue is the following: *information, new derived knowledge, plan of actions and a change of priority*. It may be that all of these types of dialogues need to be conducted to fully explore a particular topic.

Consequently, the agent needs to be able to handle nested multi-purposed dialogues with different topics. To accomplish this, the agent needs to be able to distinguish between *topic, generic goal* and have a semantic model of how these inter-relate in a particular situation. For instance, if the agent would have the dialogue with Eva instead of the nurse, the agent needs a semantic model for how walking around nighttime relates to sleep patterns, pain, cognitive ability, medication, worries, etc. (i.e., a *domain model*). Moreover, it needs strategies to plan next moves, based on a knowledge model, which may not provide a pre-defined hierarchically organized plan of actions based on goals and sub-goals to be followed, but rather a collection of prioritized actions, among which the order may become determined and changed by the dialogue evolution and Eva’s line of thinking (enabling *situatedness* of dialogues and the agent to adapt).

The *topics* relate to the human’s goals and priorities. From the software agent’s perspective, goals relate to finding answers to questions, finding and deciding about actions to make to increase levels of satisfaction, evaluate actions made, etc. Consequently, for the human, goals are purposeful, topic-driven and context dependent. Therefore, the agent in our approach combines a generic goal with a specific topic, to identify the *specific goal* with an action.

To summarize, the scenario emphasises the need for the following three semantic models for enabling human-agent dialogues: 1) a *domain model*, which contains generic knowledge about a particular domain, 2) a *user model*, which contains the collected knowledge about the human, and 3) a *dialogue activity model*, which provides the relations between topics, generic and specific goals and actions to be made by the

participating agents in a dialogue. Moreover, the third model can be seen as the generic behaviour knowledge model of the Coach Agent in dialogues. The conceptual model of a dialogue activity has been developed as a part of this work, and is presented in the following section.

4 A CONCEPTUAL MODEL OF THE DIALOGUE ACTIVITY

As described in Sect. 3, the software agent needs to share a common semantic model with the human, to be able to reason and decide upon which actions are valuable to the human. The activity-theoretical model of human activity was used for organising actions and their goals at different levels [7]. Activity Theory captures the complexity in human activity, including human *needs and motives* as driving force, *goal-directed actions*, and *operations*, which constitute basic actions conditioned by the agents and the environment.

The identified operations relate to passing information to and receive information from the other agent(s) (*send* and *receive* in Fig. 1), and wait for responses.

In addition to these basic and top levels of activity, a set of potential generic sub-actions have been defined, which may take place in the conduction of a dialogue. These are also organized in a hierarchical representation, since an action may serve as a sub-action to more than one action at different levels. However, we identify the following generic set of actions at the highest level: *information-seeking, inquiry, deliberative, and persuasive*, related to the common goals of different types of dialogues described in Sect. 3: seek information, find new knowledge, decide upon actions to be made and lead the other agent to change opinion. To these actions, we add the action *organize dialogue*, which contains the sub-actions typical for multi agent dialogues, e.g., *open* and *close*. These sub-actions are called *moves* in multiagent literature [2]. We extend the set of moves (i.e., sub-actions) and include the following moves as valid actions for the agent to take as part of the different actions: *open, close, pause, resume, ask, assert, affirm, inform, remind* and *alert*.

The higher-level actions, which relate to different types of dialogues, can be nested to meet sub-goals in the process of achieving the overall motive for the dialogue activity defined by the topic. In addition, each nested dialogue is initiated by posing a topic, in the same way as the main dialogue is initiated. As a consequence, the execution of the dialogue body needs to handle the different types of dialogues, motives, their outcomes, the organisation of these in the dynamic way, which is needed for the agent to be adaptive and flexible. In this process, some constructs are useful for formalising the dialogues and their outcomes, and for organising the process of reaching decisions about e.g, actions to make, and their reasons (arguments).

4.1 Argumentation and Types of Dialogues

The purpose of *argumentation dialogues* is to collaboratively compare different views and generate conclusions about which

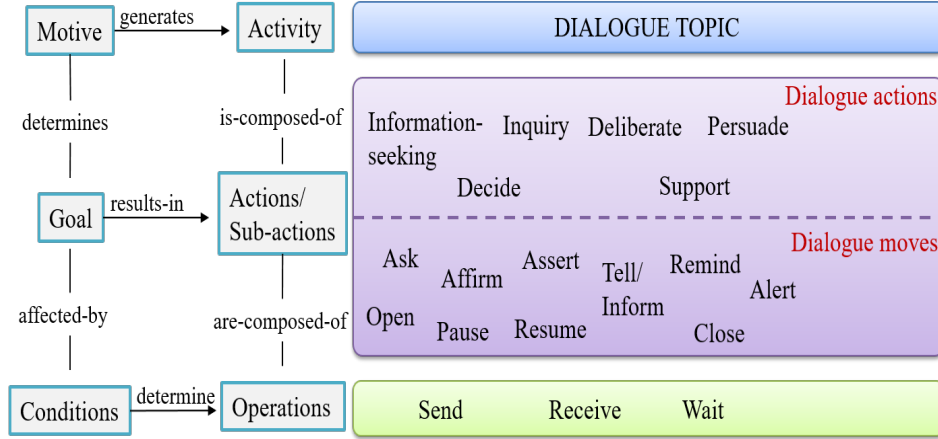


Figure 1: A conceptual model of the dialogue activity, which can be used for representing a dialogue from both a human and a software agent’s perspectives.

one is best in a situation. This can be what conclusion to draw, what new knowledge do derive, what action to make, and what changes of priority, or beliefs can be achieved. This is a decision-making approach, which provides strategies for handling conflicting information and views. Information, inference rules and conclusions are typically considered *defeasible*, which means that they can be challenged (*attacked*) and defeated. In our work, we assume that all information is defeasible, and no rules are strict rules, valid in all circumstances. Arguments can be attacked in three ways: on their *premises*, on their *inference*. In argumentation literature the notion of *argument scheme* is applied for providing semi-formal or formal templates and defeasible inference rules for different kinds of dialogues [15]. A common example is the scheme “Argument from a position to know”, which is the starting point for both the human and software agents in our work. Consequently, the agents are considered equally knowledgeable when evaluating their arguments.

The topic of dialogues is retrieved in our approach from a domain ontology, which incorporates both the Argument Interchange Format (AIF) [3] and the domain knowledge. This domain ontology is part of the ACKTUS architecture and used in the implementation of the dialogue system [12]. The domain ontology contains a concept-node system with concepts and their relations. The information-seeking dialogues have a concept as topic, which makes this type of dialogue very generic. Moreover, the domain ontology contains a kind of information node, which has a concept but no values, and functions as the topic for the supportive dialogues. We treat these as *statements* in the dialogue, and *literals* in the logic implemented in the dialogue system.

AIF distinguishes between *i-node*, a kind of information node, which in this approach is associated to both a concept and a value, and which form statements; and *s-node* (scheme-node), from which rules are generated. These AIF concepts are used as topic in inquiry, deliberation and persuasion dialogues, with some differences. The formal distinction

made in this approach between the inquiry and deliberation dialogues is that the *claim*, i.e., *i-node*, is in deliberation dialogues associated to a concept related to the node *activity* and *participation* in the domain ontology, while in the inquiry dialogues the concept is anything else than a node within the activity and participation ontology. Persuasive dialogues can have both kinds, since the focus is the *evaluation* of the phenomenon represented by the concept. This evaluation is represented by its *value* (a measurement of preference or confidence), which is targeted to be changed.

The rules together with defeasible facts (statements) fulfilling premises of the rules are used for building arguments of the form (x, a), where a is the claim and x is the support for the claim, consisting of rules and facts. An argument is posed in the dialogue with an assert move.

In our dialogue system attacks by the human on arguments are identified by a disagreeing response given by the human to the question posed by the agent using the believe move. This disagreement is respected, and the argument is considered defeated. If the human supports the argument, the argument is considered accepted and validated. The two arguments in the examples both address actions to address a problem, and as such they are not in conflict with each other. In the health domain we describe in this paper arguments are often not in direct conflict with each other. They may however support decisions or actions, which are beneficial, or important to different degrees. Moreover, as in the example, a defeated argument may be interesting in the particular situation where it was defeated, since it may provide information, which needs to be used for resolving a problem. Consequently, the state-of-the-art approaches to formal argumentation in research literature where one single winning argument is to be identified are insufficient for complex healthcare situations, and appears as less useful for a person in daily activities to be supported by a digital coach.

5 CONCLUSIONS AND FUTURE WORK

The purpose of this work was to define a generic model of purposeful human-agent dialogues about health-related topics. This was done based on analyses of a scenario, persona and models of human behaviour defined by Activity theory and argumentation theory. The major contribution of this work is the conceptual model of a dialogue activity that can be shared between the human and software agents.

Future work includes the implementation of the dialogue model together with a purposefully designed user interface, and conducting user studies with both experts in the knowledge domain and older adults. Main focus will be on the adaptability of the dialogues and their purposes to the human's line of reasoning. Future work includes also the development of heuristics for decision making that the agent can use for internal reasoning for adaptation and collaborative decision making, and the full definitions of the formal argumentation framework applied in our dialogue system.

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